

Summarized chain of thought (Ordinary two-point correlations of multiplicative functions)

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Part I considers ordinary two-point correlations of bounded multiplicative functions under uniform non-pretentiousness. Part II considers affine Liouville correlations, with the stronger quantitative target of a logarithmic-power saving at every scale. Both arguments use weighted divisor graphs, but the quantitative target requires additional control of the losses at each stage.

Part I: Qualitative corrected Elliott correlations

Original prompt (excerpts)

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Let  $\mathbb{N}=\{1,2,\ldots\}$  and  $\mathbb{D}=\{z\in\mathbb{C}:|z|\leq 1\}$ . A function  $f:\mathbb{N}\rightarrow\mathbb{D}$  is multiplicative if

$$f(mn)=f(m)f(n)\quad ((m,n)=1).$$

For a Dirichlet character  $\chi$ ,  $t\in\mathbb{R}$ , and  $(N\geq 2)$ , write

$$D(f,\chi n^{it};N)^2 = \sum_{p\leq N} \frac{1-\operatorname{Re}\bigl(f(p)\overline{\chi(p)}p^{it}\bigr)}{p},$$

where the sum is over primes  $p$ ,  $p^{it}=e^{it\log p}$ , and  $(D)$  denotes the nonnegative square root.

Say that  $f$  is uniformly non-pretentious if, for every fixed Dirichlet character  $\chi$ ,

$$\inf_{|t|\leq N} D(f,\chi n^{it};N)\rightarrow\infty\quad (N\rightarrow\infty).$$


Conjecture / Open Problem: Binary corrected Elliott conjecture for ordinary averages.

Let  $(f_1,f_2:\mathbb{N}\rightarrow\mathbb{D})$  be multiplicative functions. Suppose that at least one of  $(f_1,f_2)$  is uniformly non-pretentious. Then for every fixed pair of distinct shifts  $(h_1,h_2\in\mathbb{Z}_{\geq 0})$ ,

$$\frac{1}{N}\sum_{n=1}^N f_1(n+h_1)f_2(n+h_2)=o(1)\quad (N\rightarrow\infty).$$

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1 From the single-scale obstruction to weighted divisor graphs

Reading Klurman, Mangerel, and Teräväinen [6], the assistant matched the prompt's uniform hypothesis to Tao–Teräväinen's cancellation outside exceptional scales [12]. It distinguished both that conclusion and Tao's logarithmically averaged theorem [13] from ordinary convergence at every scale. The Liouville specialization already contained ordinary two-point Chowla.

Attempts to build counterexamples through prime phases encountered dense compatibility constraints. Proof proposals instead extracted prime divisors and encoded neighboring correlations in arithmetic graphs. Their central difficulty was removing sparse divisibility conditions without changing the scale. Comparable-size primes supplied insufficient reciprocal mass; broader dilations mixed different scales. The assistant examined Tao’s entropy decrement argument [13], whose independence estimate selects an unspecified scale, and compared its graph proposals with the prime-divisibility expansion method of Helfgott and Radziwiłł [4].

The assistant next explored positive kernels combining a real scale with congruence information in finite adeles. Smoothing the real coordinate appeared promising, but the assistant found that restoring the finite coordinate prevented it from inferring the required continuity. It considered the equilibrium-state classifications of Laca–Raeburn and Laca–Neshveyev [7, 8], but identified a missing tracial property closely related to the desired cancellation. Neshveyev’s adelic ergodicity theorem [10] suggested a dilation-invariant formulation; the assistant found that infinite measure blocked the simple invariant-probability argument and that normality remained an extra assumption. It considered the mixing-subalgebra framework of Cameron, Fang, and Mukherjee [2], while observing that normal mixing results did not classify the singular kernels it needed. It also inspected Tao–Teräväinen’s quantitative correlation theorem [14]: the permitted shifts and moduli were too small for its proposed dilation window.

Helfgott’s work on homogeneous polynomial parity and root numbers [5] suggested another direction. The assistant tried a polynomial substitution preserving Liouville signs, but its sparse image left it without a positive-density argument.

Returning to graphs, the attempt used Pilatte’s *Improved bounds for the two-point logarithmic Chowla conjecture* as a model for centered divisibility operators [11]. Weighting squarefree divisors by $A^{\omega(d)}$ produced the broad degree $(1 + A)^{\omega_S(n)}$, where ω_S counts prime factors from a selected set. With $\beta = (1 + A)^{-1}$, centering at β^2/p appeared to reduce expansion losses enough for the short exponential-sum estimates of Matomäki, Radziwiłł, and Tao [9] to help. It studied Pilatte’s fourth-moment uncentering argument and Tao–Teräväinen’s extensions of the decoupling estimate [11, 14] as models for this step. The assistant then returned to the estimates needed for this parameter choice. Localizing divisor products to narrow intervals required uniform concentration along long graph walks; repeated primes could retain too much dependence. It proposed smooth cutoffs and triangular divisibility constraints, while continuing to ask how nonconsecutive occurrences of a prime could supply enough independent constraints. These localization and dependence questions guided the next stage of its weighted-graph argument.

2 Preserving cancellation while auditing the divisor graph

The corrected Elliott attempt developed and then audited a weighted graph intended to prove ordinary two-point cancellation for multiplicative functions of modulus at most one under the corrected nonpretentiousness hypothesis. The assistant used Pilatte’s work on logarithmic Chowla for models of triangular divisibility constraints, prohibited paths, and combinatorial sieves, then investigated how to adapt them to the weighted graph [11]. The conditional sieve was based on the combinatorial sieve of Helfgott and Radziwiłł [4].

Squarefree composite steps were restricted to a narrow interval near a scale H . Normalizing prime-product weights by vertex degrees changed the centering required to cancel primes appearing on just one trace edge. Fresh prime variables were intended to yield a reciprocal logarithmic-scale saving per step, while divisor-count cutoffs compensated steps lacking such variables. Repeated vertices, disconnected prime occurrences, and branching paths complicated this calculation.

VERBATIM EXCERPT

Singleton cancellation requires n residue for singleton primes approximate independent uniform when no Y failure, even conditioned on repeated prime hits etc. Need upper bound for deviation.

The assistant identified a failure in its return-edge argument: bounding a return factor pointwise and removing it before signed integration could destroy the cancellation needed elsewhere. The repair retained return factors during primewise integration. A further revision separated heavily weighted, uncentered core primes C from centered marks Z , using

$$D(n) = \beta^{-\omega_C(n)} 2^{\omega_Z(n)}, \quad \beta = (1 + A)^{-1}.$$

Here $\omega_S(n)$ counts prime divisors from S ; marks were centered at $1/(4p)$, and cutoffs depended only on core primes. Conditional sieve expansions and tree decompositions were proposed to control damaged singleton cancellations, but expanding the sieve could itself restore prohibited paths.

For the uncentering step, it examined Pilatte's fourth Fourier moment estimate and its sieve input from Friedlander and Iwaniec's *Opera de Cribro*, then proposed a rough-number analogue [11, 3].

The assistant then checked the parameters and the short exponential-sum theorem of Matomäki, Radziwiłł, and Tao as an input to removing the centering terms [9]. Their corrected nonpretentiousness condition requires the minimum pretentious distance over characters of bounded modulus and twists $|t| \leq X$ to diverge for every fixed modulus bound. Excluding each fixed twist separately is insufficient. The assistant clarified this premise and organized the candidate proof around further checks of sieve independence, trace enumeration, and the proposed spectral estimate. In revisiting earlier approaches, the assistant retained Tao and Teräväinen's almost-all-scales cancellation, presented as Theorem A by Klurman, Mangerel, and Teräväinen, as a benchmark that still allowed exceptional scales [12, 6]. It also recalled the quantitative correlation theorem of Tao and Teräväinen: its uniform polylogarithmic range of shifts and moduli was too short for the rational configurations the assistant had considered [14].

3 A two-band graph and the corrected-Elliott conclusion

The corrected-Elliott attempt ends by refining its contradiction argument for a persistent ordinary correlation of multiplicative functions of modulus at most one, with at least one uniformly nonpretentious. The assistant continued its graph construction, sieve truncation, and uncentering argument from the preceding sections. The refinement assigns distinct roles to two prime bands. Heavily weighted core primes pay for narrow multiplier intervals and return steps in closed walks; center primes supply signed cancellation through factors $1_{p|n} - 1/(4p)$.

A simplification replaces layered core bands by one narrow upper band. First visits in each closed walk form a rooted tree. Connected prime-label subtrees receive weights whose sum is intended to cancel the background normalization. Fresh core primes supply interval savings, while exceptional residue coincidences require separate counting. The argument repeatedly tests repeated edges, missing center factors, shared divisors, and interactions among cutoff penalties.

The assistant found no fatal contradiction in these tests, but continued to question how to preserve singleton cancellation while conditioning on forbidden paths. The proposed repair uses truncated inclusion-exclusion and triangular divisibility constraints, with factorial denominators retained when summing unordered prime assignments.

The assistant used Pilatte's *Improved bounds for the two-point logarithmic Chowla conjecture* [11] as a model for high-trace estimates, prohibited progressions, and triangular constraints controlling the combinatorial-sieve remainder. For uncentering it used Theorem 1.7 of Matomäki, Radziwiłł, and Tao's *An averaged form of Chowla's conjecture* [9], checking that finite-prime cutoff twists preserve the needed

nonpretentiousness as the averaging scale tends to infinity. It combined this short-exponential-sum input with rough-number sieve and fourth-moment bounds, taking the averaging limit before the prime parameter grows. In the closing derivation it replaced an appeal to an upper-bound sieve by truncated inclusion–exclusion estimates for one and two rough integers. The assistant concluded that sufficiently small centered-graph and uncentering bounds would contradict the lower bound transferred from a persistent correlation.

The claimed cancellation depends on the conditional-integration, exceptional-pattern enumeration, triangular-extraction, and short-interval estimates proposed along this route. In particular, the conditional singleton-cancellation and sieve-remainder bounds are essential to the conclusion.

Part II: Quantitative affine Chowla correlations

Original prompt (excerpts)

Let $\lambda(n)=(-1)^{\Omega(n)}$, where Ω counts prime factors with multiplicity. Prove that an absolute $c>0$ exists such that, for every fixed pair of affine forms $L_j(n)=a_jn+b_j$ with positive integers a_j , nonnegative integers b_j , and $a_1b_2-a_2b_1\neq 0$,

$$|\sum_{n\leq X}\lambda(L_1(n))\lambda(L_2(n))|\ll_{\{a_1,a_2,b_1,b_2\}}X/(\log X)^c$$

for all $X\geq 3$. The sum is unweighted and the estimate must hold at every scale.

4 A quantitative affine Chowla attempt

Pilatte’s logarithmic Chowla work [11] suggested using many prime families. Pilatte attributed the combinatorial sieve to Helfgott and Radziwiłł [4]. The assistant examined repeated prime coordinates in composite steps and the comparison between finite intervals and independent residues:

VERBATIM EXCERPT

Need exact cancellation singleton and nonnegative support of low-rank. Finite CRT prevents mixing over P for cutoffs.

Here CRT means the Chinese remainder theorem. The combined residue modulus could exceed the interval length. The assistant explored Braverman’s bounded-independence theorem for Boolean circuits [1] as a possible way to transfer independent-residue calculations to finite intervals. Applying it would require a suitable circuit encoding of the deletion predicates and control of the available degree of independence.

At the combinatorial turning point, the assistant sought compatible deletion witnesses for every prime label occurring only once:

VERBATIM EXCERPT

Mixed difference property stronger than pairwise changes: if total mixed difference nonzero, expand product over **all** potential witness exclusion booleans. Then at least one intersection of witnesses in SAME hybrid covers ALL singleton labels; otherwise expansion cancels per term. hybrid deterministic choice finite, no counting huge.

The proposed common-residue witness cover would supply triangular divisibility constraints, while compressed composite walks needed a new low-rank forest encoding. The assistant continued to seek a uniform encoding cost for repeated and omitted prime labels.

In examining Pilatte’s fourth-moment argument, the assistant encountered its classical sieve input from Friedlander and Iwaniec’s *Opera de Cribro* [3] and its elementary mean-value estimate attributed to Tenenbaum [17]. Removing the centering terms prompted a proposed fourth Fourier moment bound for rough-number weights and examination of short-interval Liouville estimates of Matomäki, Radziwiłł, and Tao [9]. Cutoffs and deletion errors still threatened the saving. The assistant carried witness compatibility, finite-interval transfer, and the combined parameter losses forward as estimates it still needed to control.

5 Closing the quantitative affine Chowla argument

The construction reduced affine forms to a fixed-shift correlation restricted to a residue class. It combined products of primes from separated ranges with weighted squarefree padding and narrow logarithmic intervals. A proposed nonbacktracking matrix prohibited consecutive repetition of the entire tuple of centered primes. Its trace expansion separated configurations with many mean terms, high constraint rank, or many singleton prime labels. The singleton case adapted Pilatte’s prohibited-sequence and triangular-system ideas [11], using jointly present witnesses and triangular congruence elimination; the remaining patterns were encoded as forests. Prime-counting estimates in fixed residue classes, checked against Tao’s notes [15], controlled the prime supplies and padding distribution.

For the finite-average comparison with independent residues, the assistant checked Braverman’s bounded-independence theorem [1]. For removing the centering terms, it checked corrections to the Matomäki–Radziwiłł–Tao short exponential-sum input [9]. It combined these inputs with proposed fourth-moment estimates, a padding cutoff, and a spectral passage from the nonbacktracking matrix to the centered adjacency sum. The literature check also consulted Tao’s discussion of his paper with Teräväinen, *Quantitative correlations and some problems on prime factors of consecutive integers* [14, 16], which distinguished logarithmic averaging and almost-all-scale conclusions from the all-scale target.

With $\delta = 1/200$, $L = (\log X)^{1/A}$ and $J = \lfloor \delta \log L / (6W) \rfloor$, the assistant concluded that sufficiently large absolute constants A, W made the normalized fixed-shift correlation

$$F(X) = X^{-1} \sum_{n \leq X} \mathbf{1}_{n \equiv b \pmod{l}} \lambda(n) \lambda(n+h)$$

decay like e^{-J} . Here $\eta = e^{-J}$, $K = e^{4J}$, and h, l are the fixed shift and modulus:

VERBATIM EXCERPT

With (J, η, K) as above the right side is $O_{h,l,W}(e^{-J})$. Thus the desired exponent for $F(X)$ can be $(c = \delta/(6WA) > 0)$, with no discarded scales.

Finally, the assistant returned to the affine forms using $l = a_1 a_2$, $b = \min(a_2 b_1, a_1 b_2)$, and $h = |a_1 b_2 - a_2 b_1|$. This reduction would give the all-scale affine bound with an absolute positive exponent, provided the proposed simultaneous-witness elimination, forest-counting, finite-residue comparison, and spectral-transfer bounds hold with the stated uniform losses.

References

- [1] Mark Braverman. *Polylogarithmic independence fools AC^0 circuits*. Journal of the ACM **57**(5) (2010), 28:1–28:10. <https://doi.org/10.1145/1754399.1754401>.

- [2] Jan Cameron, Junsheng Fang, and Kunal Mukherjee. *Mixing subalgebras of finite von Neumann algebras*. New York Journal of Mathematics **19** (2013), 343–366. <https://nyjm.albany.edu/j/2013/19-17.html>.
- [3] John Friedlander and Henryk Iwaniec. *Opera de Cribro*. American Mathematical Society, Colloquium Publications 57 (2010). <https://bookstore.ams.org/coll-57>.
- [4] Harald Andrés Helfgott and Maksym Radziwiłł. *Expansion, divisibility and parity*. arXiv preprint arXiv:2103.06853v2 (2021). <https://arxiv.org/abs/2103.06853v2>.
- [5] Harald Andrés Helfgott. *Root numbers and the parity problem*. Ph.D. thesis, Princeton University (2003). arXiv:math/0305435. <https://arxiv.org/abs/math/0305435>.
- [6] Oleksiy Klurman, Alexander P. Mangerel, and Joni Teräväinen. *On Elliott’s conjecture and applications*. Proceedings of the London Mathematical Society **133**(3) (2026), e70205. Preprint version: arXiv:2304.05344v2 (2023). <https://doi.org/10.1112/plms.70205>.
- [7] Marcelo Laca and Iain Raeburn. *Phase transition on the Toeplitz algebra of the affine semigroup over the natural numbers*. Advances in Mathematics **225**(2) (2010), 643–688. <https://doi.org/10.1016/j.aim.2010.03.007>.
- [8] Marcelo Laca and Sergey Neshveyev. *Type III_1 equilibrium states of the Toeplitz algebra of the affine semigroup over the natural numbers*. Journal of Functional Analysis **261**(1) (2011), 169–187. <https://doi.org/10.1016/j.jfa.2011.03.009>.
- [9] Kaisa Matomäki, Maksym Radziwiłł, and Terence Tao. *An averaged form of Chowla’s conjecture*. Algebra & Number Theory **9**(9) (2015), 2167–2196. Corrected version: arXiv:1503.05121v3 (2022), <https://arxiv.org/abs/1503.05121v3>. <https://doi.org/10.2140/ant.2015.9.2167>.
- [10] Sergey Neshveyev. *Ergodicity of the action of the positive rationals on the group of finite adeles and the Bost-Connes phase transition theorem*. Proceedings of the American Mathematical Society **130**(10) (2002), 2999–3003. <https://doi.org/10.1090/S0002-9939-02-06449-3>.
- [11] Cédric Pilatte. *Improved bounds for the two-point logarithmic Chowla conjecture*. Journal of the American Mathematical Society (2026). Published online 10 September 2026. Cited version: arXiv:2310.19357v3, <https://arxiv.org/abs/2310.19357v3>. <https://doi.org/10.1090/jams/1084>.
- [12] Terence Tao and Joni Teräväinen. *The structure of correlations of multiplicative functions at almost all scales, with applications to the Chowla and Elliott conjectures*. Algebra & Number Theory **13**(9) (2019), 2103–2150. <https://doi.org/10.2140/ant.2019.13.2103>.
- [13] Terence Tao. *The logarithmically averaged Chowla and Elliott conjectures for two-point correlations*. Forum of Mathematics, Pi **4** (2016), e8. <https://doi.org/10.1017/fmp.2016.6>.
- [14] Terence Tao and Joni Teräväinen. *Quantitative correlations and some problems on prime factors of consecutive integers*. arXiv preprint arXiv:2512.01739v2 (2026). First version 2025. <https://arxiv.org/abs/2512.01739v2>.
- [15] Terence Tao. *254A, Notes 2: Complex-analytic multiplicative number theory*. Lecture notes on *What’s new* (2014). 9 December 2014. <https://terrytao.wordpress.com/2014/12/09/254a-notes-2-complex-analytic-multiplicative-number-theory/>.
- [16] Terence Tao. *Quantitative correlations and some problems on prime factors of consecutive integers*. Blog post on *What’s new* (2025). 1 December 2025; discussion of the joint paper with Joni Teräväinen. <https://terrytao.wordpress.com/2025/12/01/quantitative-correlations-and-some-problems-on-prime-factors-of-consecutive-integers/>.

- [17] Gérald Tenenbaum. *Introduction to analytic and probabilistic number theory*. American Mathematical Society, Graduate Studies in Mathematics 163 (2015). Third edition; translated by Patrick D. F. Ion. <https://www.ams.org/books/gsm/163/>.